

User Guide — Nanotec Inspection-Table Controller

This is the operator guide for the multi-axis motion application that drives the inspection table's four EtherCAT axes — **X, Y, Z, and Θ (the rotary chuck)** — through Nanotec drives using **NanoLib** over **EtherCAT (CoE / CiA 402)** with an **Npcap soft master**.

Status: This application is still being brought up on real hardware. Treat every first motion on a new machine as a commissioning step: keep the E-stop within reach, start at low jog speeds, and confirm each axis moves the way you expect before trusting automated moves (Home All, Go Home, Find Limits).

For how the software works internally, see the **Developer Guide** (Developer Guide/README.md).

1. Before you start

Software the PC needs

- **Npcap** — installed with "**Install Npcap in WinPcap API-Compatible Mode**" ticked. NanoLib's EtherCAT master sends raw packets through Npcap; without it, no drives are found.
- **.NET 10 (Windows) runtime / SDK** — to build and run.
- The **NanoLib** package (nanotec.services.nanolib 1.4.0) is restored automatically as a project dependency.

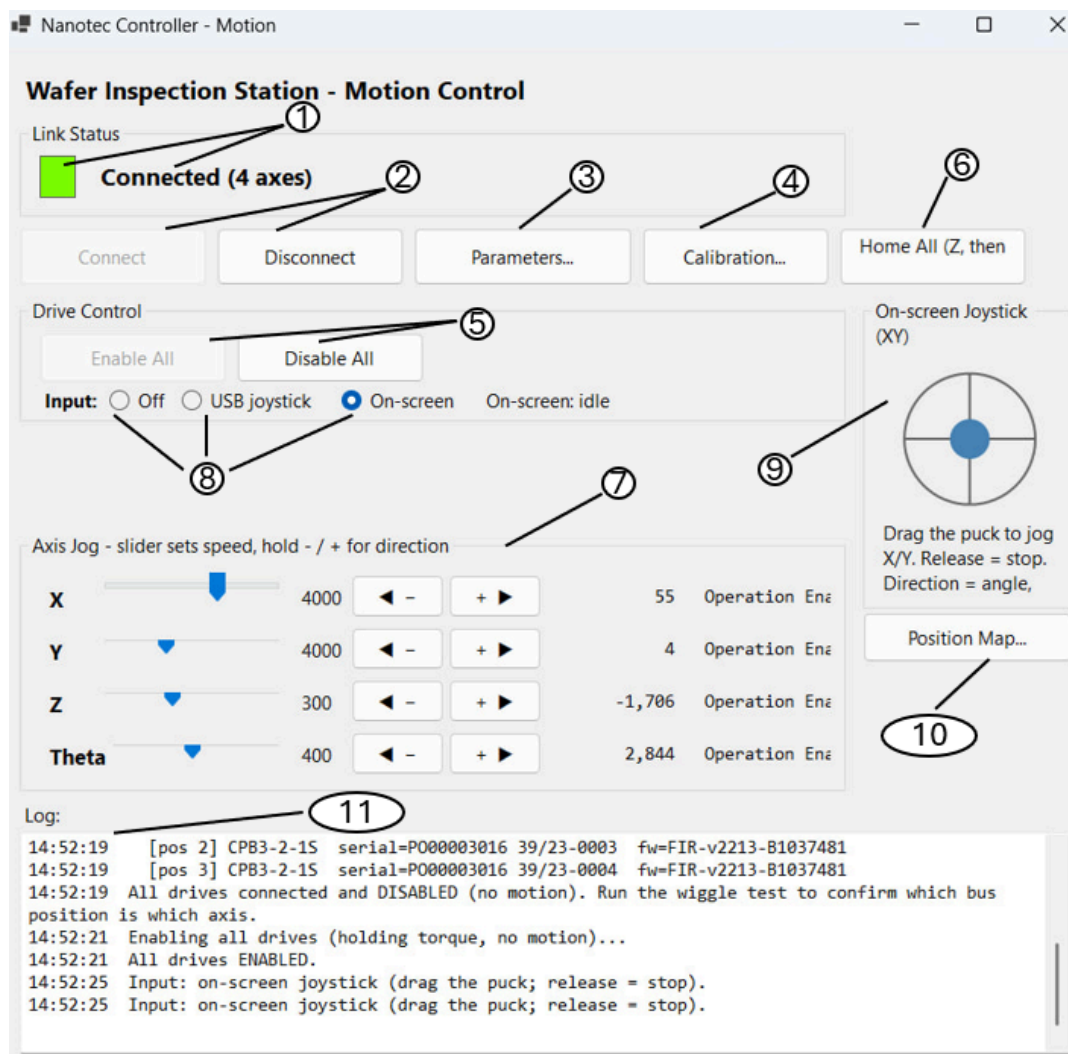
The application must run as **Administrator**

Raw packet access through Npcap requires elevation. If you launch without admin rights, the bus scan will either find no adapters or fail to open the EtherCAT NIC.

Hardware connections

1. Connect the PC's dedicated NIC straight into the **EtherCAT IN** port of the first drive in the chain.
 2. Power the drive stage from its external supply.
 3. The four drives are daisy-chained. Their **bus order is fixed as X, Y, Z, Θ** — i.e. the first drive on the line is X, the last is Θ .
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2. The main window at a glance



Ref. no	Area	What it does
1.	Connection LED + status	Red = disconnected, green = connected, amber = busy (an operation is running).
2.	Connect / Disconnect	Open or close the link to all drives.
3.	Parameters...	Opens the parameters window: a read-only dump of each drive's limits & unit/scaling settings, plus an expert option to write objects (RAM) or save to NV (see §10).
4.	Calibration...	Opens the travel-limits & Home window (see §8).
5.	Enable All / Disable All	Energise / de-energise all drives.

6.	Home All	Retract Z, then send X & Y to their home positions (see §7).
7.	Per-axis rows (X / Y / Z / Θ)	Speed slider, -/+ hold-to-jog buttons, and a live position + state readout per axis.
8.	Input source (Off / USB / On-screen)	Selects the joystick input (see §6).
9.	On-screen joystick puck	Drag-to-move analog joystick for X/Y.
10.	Position Map...	Opens the absolute-positioning window: click an XY grid (or type X/Y/Z) to set a target, then Go (see §9).
11.	Log	Timestamped record of everything the app did. Read it — it reports every stop, limit, and error.

3. Connecting

1. Click **Connect**.
2. The app scans the PC's network adapters and shows a **bus picker** dialog. EtherCAT- capable adapters are tagged [**EtherCAT**]. Choose the one your drives are on and click **Connect**.
3. The app connects to **every drive on the line in bus order** and logs each drive's axis name, serial number, and firmware. Confirm it found **4 drives** and that the serials line up with the axes you expect.

Connecting never moves anything. All drives come up **disabled** (no torque, no motion). This is deliberate — bring-up is always a separate, explicit step.

If the app finds a different number of drives than expected, it warns you in the log. If the axis mapping can't be completed (e.g. a drive is missing), it disconnects rather than run with a partial table.

4. Enabling the drives

Click **Enable All**. The app walks every drive through its power-on sequence and leaves each one **holding position with zero commanded speed** — energised but not moving.

- The status row for each axis should read **Operation Enabled**.
- **No axis should move when you enable.** If one does, disable immediately and report it — that indicates a leftover motion target or a running on-drive

program.

Disable All stops and de-energises every axis. Switching the input source to **Off** or disabling always halts motion first.

5. Jogging with the on-screen buttons

Each axis row has:

- A **speed slider** — that axis's jog speed (in the drive's own velocity units). The live value is shown beside it. Each axis has its own sensible default and maximum.
- **- and + buttons** — **hold to move, release to stop**. Motion only lasts as long as you hold the button. There is no "latched" jog.

The speed is taken at the moment you press, so set the slider first, then press and hold.

6. Joystick control

Pick the input with the **Off / USB / On-screen** radio buttons (only available once drives are enabled). The two joysticks are mutually exclusive.

USB joystick

Any controller Windows lists in **joy.cp1** works. The default button map:

Button	Function
1	Deadman — <i>must be held</i> for any motion. Release = immediate stop.
2	Fast — multiplies the jog speed (capped at each axis's slider maximum).
3 / 4	Z - / Z +
5 / 6	Θ - / Θ +

The main stick drives **X/Y**; the D-pad/hat also works. Stick directions are digital (full speed when deflected). **If the joystick is unplugged or vanishes mid-use, all axes stop.**

On-screen joystick (mouse)

Drag the **puck** inside the circle. Unlike the USB stick this is **analog**: the puck's angle sets the X/Y direction and how far you push sets the speed (rim = that axis's slider

speed). **Release the mouse and the puck springs back to centre → motion stops.**
There is no deadman — holding the mouse *is* the intent.

7. Soft travel limits (automatic protection)

Once you've calibrated an axis's **Min/Max** (see §8), the app watches each axis while you jog and **stops it if it tries to travel past a stored limit**. You can always jog **back into range** — only further-out motion is blocked.

Important caveats:

- This is a **software** guard polled a few times a second, so expect a little overshoot at high speed. Where physical limit switches exist, **they** are the real safety; the soft limit is a convenience guard.
 - On this machine, **X's + end and both ends of Z have no working limit switch**, so the soft limit is the *only* protection there. Calibrate those axes before jogging them far, and keep speeds modest.
 - If `calibration.json` is missing or unreadable at startup, the app logs a **"starting with NO soft limits"** warning. Take it seriously — re-calibrate before jogging.
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8. Calibration window (travel limits & Home)

Calibration - travel limits & Home

Jog an axis to a position in the main window, then Set Min / Set Max here (Clear Min / Clear Max removes a stored limit). Home = centre of the two limits for X/Y; Z's Home is set explicitly. Find Limits drives Y into its switches automatically.

Axis	Min	Max	Home	Buttons
X	-136,142	136,253	55	Set Min, Set Max, Go Home, Clear Min, Clear Max
Y	-79,208	79,199	-4	Set Min, Set Max, Find Limits, Go Home, Clear Min, Clear Max
Z	-2,285	1,848	-1,706	Set Min, Set Max, Set Home, Go Home, Clear Min, Clear Max

Open it with **Calibration....** It shows X, Y, Z (Θ has no home and is excluded). All calibration values are saved to `calibration.json` next to the app and survive restarts.

For each axis:

- **Set Min / Set Max** — jog the axis to a position in the main window, then click to **capture the current position** as that limit.

- **Clear Min / Clear Max** — removes a stored limit (back to "none"). This is a local edit only — it moves nothing — and also drops any jog block that limit was enforcing.
- **Set Home** (Z only) — captures Z's explicit home position.
- **Find Limits** (Y only) — **automatically** drives Y into each end switch, records both edges as Min/Max, and sets Home to the centre. Watch it run; it backs off the switches when done.
- **Go Home** — moves the axis to its home (the **centre of Min/Max** for X/Y, the explicit Home for Z) and reports how close it landed.

Home model summary:

- **X / Y:** Home = midpoint of the two limits (needs both Min and Max set).
- **Z:** Home = the explicit position you captured with Set Home.

Special Note

It is **highly recommended** to set the Z-minimum above the chuck.

Home All

The **Home All** button on the main window runs a safe homing sequence:

1. **Z moves to its home first** (e.g. retracts to a safe height) — and the app **confirms Z arrived** before doing anything else.
2. **Then X and Y move to their homes together.**

It requires Home to be defined for **all three** of X, Y, Z; otherwise it refuses and tells you which are missing — so X/Y never traverse while Z is still down.

9. Position Map (go to a coordinate)

Open it with **Position Map....** It shows an **XY grid** of the table's travel envelope on the left and numeric **X / Y / Z** target fields with a **Go** button on the right.

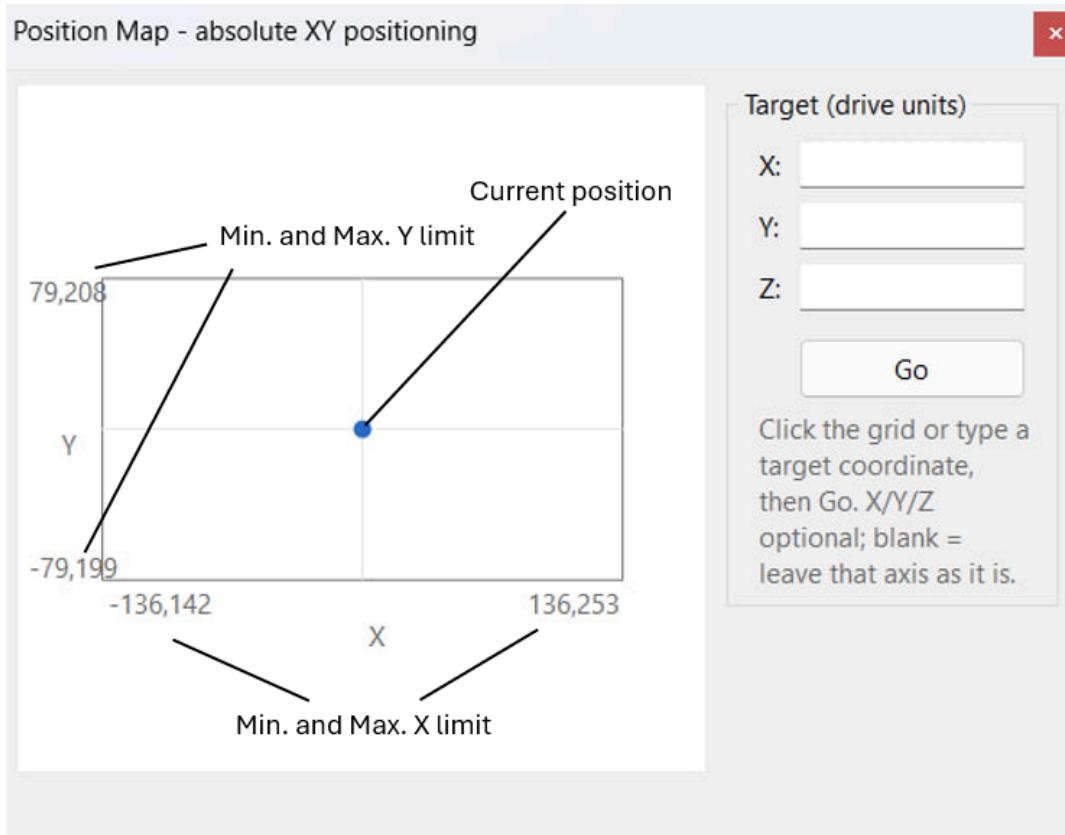
Pick a target two ways — nothing moves until you press Go:

- **Click the grid** — stages a target crosshair at that spot and fills the X/Y fields. The filled blue dot is the live current position; the hollow red crosshair is your staged target.
- **Type into X / Y / Z** — the crosshair follows what you type (Z has no grid axis, so it's numeric only).

Then press **Go** to move. The same rules as before apply:

- Any field left **blank** means "leave that axis where it is."

- Targets are **range-checked against each axis's Min/Max**. If any one is out of range, the **whole move is cancelled** and the offending value is logged.
- The entered axes move together. Values are in the same drive units shown as Min/Max.



Notes:

- The grid stays **greyed out until both X and Y limits are calibrated** (see §8) — it needs the envelope to map clicks to coordinates.
- **Z is not on the grid**. There is no automatic Z-collision check — guard it by setting Z's **Min limit above the chuck** so a too-low Z target is rejected by the range check.
- **Go** is only enabled while the drives are enabled and idle; the window can be left open while you jog from the main form to fine-tune.

10. Parameters (read & write drive settings)

Open it with **Parameters....** It's a separate window with its own output log, and it does two very different jobs.

Read Params — safe, read-only

Read Params (all axes) dumps each connected drive's key configuration to the window's log **without writing anything** — current/torque limits, max speed, the

unit/scaling objects that define what "position" and "velocity" units actually mean, and a motion-state snapshot (mode, target, profile accel/decel).

Typical use: read once, **power-cycle the drives**, read again, and compare to confirm the drives kept their settings in non-volatile memory.

Parameters - read / write drive objects

Write object (expert)

Axis: X, Object: 0x 6084, Sub-index: : 00, Value: , Bits: 32

Write, Save to NV

Value: decimal, or 0x.. for hex. Write = RAM; Save to NV = NV

Read Params (all axes)

15:30:08 Feed constant: shaft 0x6092:02 = 1

15:30:08 Velocity factor: num 0x6096:01 = 1

15:30:08 Velocity factor: den 0x6096:02 = 1

15:30:08 Motion state (read right after a Move To / Go Home):

15:30:08 Mode display 0x6061:00 = 3 (1=PP 3=PV 6=home)

15:30:08 Mode commanded 0x6060:00 = 3

15:30:08 Statusword 0x6041:00 = 0x1637

15:30:08 Target position 0x607A:00 = 1 pos units

15:30:08 Position actual 0x6064:00 = 2844 pos units

15:30:08 Profile velocity 0x6081:00 = 2000 vel units

15:30:08 Profile acceleration 0x6083:00 = 4000 vel units/s

15:30:08 Profile deceleration 0x6084:00 = 4000 vel units/s

15:30:08 === Done. Re-run after a power cycle and compare to confirm NV persistence. ===

Write object / Save to NV — expert, changes the drive

The write row sets **any** object-dictionary entry on a chosen axis. Enter the object as index : sub in hex (e.g. 6084 : 00), the value in decimal or 0x... hex, and its size in bits (8 / 16 / 32):

- **Write** — writes the value to the drive's **RAM**: it takes effect now but is lost on the next power-cycle.
- **Save to NV** — persists that axis's **current** parameter values to non-volatile memory (object 0x1010:01), so they survive a power-cycle.

Both ask for confirmation first, because there is no validation beyond the drive's own — a wrong object or value can change any writable setting. (For example, to fix X's slow stop you can write 6084 = 20000 on X, then **Save to NV**.)

The window only works while the drives are connected, and pauses live polling while it reads or writes.

11. Safety behaviours you can rely on

- **Connecting performs no motion.** Drives come up disabled.
 - **Enabling holds position** with zero speed — no lurch (provided no on-drive program is running).
 - **All jogging is momentary** — release the button, release the deadman, or re-centre the puck and motion stops.
 - **Losing window focus stops everything** and pauses the joystick. (A running operation — Home, Find Limits, Move, or Go Home — is left alone, since it owns the drives.)
 - **Losing the USB joystick stops all axes.**
 - **Soft limits stop outward jogs** on calibrated axes.
 - **Closing the window** disables the drives and disconnects cleanly.
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12. Troubleshooting

Symptom	Likely cause / fix
Connect finds no buses	Npcap not installed / not in WinPcap-compatible mode; app not run as Administrator.
Connect finds no drives	Cabling (IN vs OUT port), drive power, wrong adapter chosen in the bus picker.
Wrong number of drives found	An unpowered drive or a bad daisy-chain link — check the log for the count.
An axis moves on Enable	Disable immediately. Suspect a leftover target or an on-drive (NanoJ) program still running.
Jog buttons are greyed out	Drives aren't enabled, or an operation is busy (amber LED), or the axis is parked against a soft limit in that direction.
"starting with NO soft limits" at launch	calibration.json was missing/corrupt (a calibration.corrupt.json backup is kept). Re-calibrate.
Lost contact with a drive in the log	The link dropped after several failed reads; reconnect. The soft master is not real-time, so the occasional miss is tolerated before it gives up.

*Position and velocity values are shown in the **drive's own configured units**, not yet converted to millimetres or degrees. Until that scaling is confirmed on hardware, treat the numbers as raw drive units and verify physical magnitudes by observation.*